Healthcare Strategy Platform — Discussion & PRD History

Complete transcript of system architecture, multi-agent specifications, PRD, and slide design guidelines.

USER

I will be building a working prototype for a healthcare strategy team. The project goal is a tool that does 4 things:

1. generates a structured brief for a given medical condition
2. states the current standard of care along with mechanism of action
3. provides an overview of emerging treatments in development (including clinical trials and basic research) along with how they differ from the current standard of treatment
4. provides an overview of key companies, institutions, researchers, etc.

For a tool that completes all of the above can you state what components are needed for the input and what is needed for the output. For example, do we need something that needs a particular patient's lab work, medical history etc. Additionally what key features need to be provided in the output brief.

ASSISTANT

To build a working prototype for a healthcare strategy team, your architecture will shift away from individual patient-level data. The tool requires **population-level, epidemiological, and biomedical data inputs**.

Part 1: System Inputs Needed

- **Condition Identifier (Core Trigger):** Medical term, ICD-10 code, or MeSH term (e.g., *"Plaque Psoriasis"*).
- **Analysis Scope & Filters:** Geography/Market, Development Phase Filter, Time Horizon.
- **External Data Connectors:** ClinicalTrials.gov API, PubMed/MEDLINE, Patent/Grant databases.

Part 2: Key Features of Output Brief

- **1. Structured Brief:** Epidemiology, Disease Pathophysiology, Unmet Needs Matrix.
- **2. Current Standard of Care (SoC) & MoA:** First vs. Later-Line Therapies, Mechanism of Action, Efficacy/Safety Benchmarks.
- **3. Emerging Treatments:** Pipeline by Phase, Modality Classification, Differentiation Matrix, Clinical Trial Signals.
- **4. Key Stakeholders:** Industry Sponsors, Academic Institutions, Key Opinion Leaders (KOLs).

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I want to edit the output in these ways:

1- I want the "Structured Brief for the Medical Condition" and "Current Standard of Care (SoC) & Mechanism of Action (MoA)" to be completed by one agent. When generating this information search through pubmed, relevant peer-reviewed research papers, etc to draft the brief. Use figures if necessary.

2- For "Emerging Treatments in Development (Pipeline Analysis)" perform a comprehensive analysis of clinicaltrials.gov, pubmed, etc. Show current trials, their respective status. Additionally, create a subsection within this section that shows clinical trials and associated companies/institutions that have not succeeded or have not met endpoints.

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Updated specification incorporating a multi-agent orchestration architecture:

- **Agent 1 (Clinical & SoC Specialist):** Drafts Section 1 (Condition Brief) and Section 2 (Standard of Care & MoA) using PubMed, MEDLINE, and clinical guidelines.
- **Agent 2 (Emerging Pipeline & Failure Analysis Specialist):** Drafts Section 3 (Pipeline & Failure Analysis) and Section 4 (Stakeholders) querying ClinicalTrials.gov, PubMed, and SEC filings. Includes a dedicated *Failed / Terminated / Unmet Endpoint Subsection*.
- **Aggregator Agent:** Combines outputs into a unified executive brief.

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generate a comprehensive PRD of these agents and modules

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Generated full Product Requirement Document (PRD) detailing Agent Specifications, Data Pipelines (PubMed E-utilities, ClinicalTrials.gov API v2), Module Specifications (Condition Brief, SoC Benchmarks, Active Pipeline, Failed Trials Audit, Stakeholder Maps), Technical API Matrices, and Non-Functional Requirements.

USER

don't generate the slide only tell me the components of the slide and where everything will appear on the slide. this slide must be designed for non-technical audiences

ASSISTANT

Executive Slide Blueprint for Non-Technical Audiences:

- **Top Section (Problem vs. Solution):** Side-by-side cards comparing manual research friction vs. multi-agent automation.
- **Middle Section (3-Step Schematic Flow):** Left-to-right visual sequence: User Input → Specialized AI Agents (Agent 1 & Agent 2 in parallel) → Aggregator Agent & Final Executive Brief.
- **Bottom Section (Data Sources & Trust Layer):** Visual badges highlighting official integrations with PubMed, ClinicalTrials.gov, and Regulatory feeds to explain how hallucinations are avoided.